



Radical embodied cognitive science and “Real Cognition”

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Abstract

A persistent criticism of radical embodied cognitive science is that it will be impossible to explain “real cognition” without invoking mental representations. This paper provides an account of explicit, real-time thinking of the kind we engage in when we imagine counter-factual situations, remember the past, and plan for the future. We first present a very general non-representational account of explicit thinking, based on pragmatist philosophy of science. We then present a more detailed instantiation of this general account drawing on nonlinear dynamics and ecological psychology.

Keywords Embodied cognition · Thinking · Representation · Artifactualism · Resonance · Ecological psychology

1 Introduction

Rodney Brooks’ landmark article “Intelligence without representation” (1991) was published along with a partly sympathetic critique by David Kirsh “Today the Earwig, Tomorrow Man” (1991). Kirsh’s title makes his line of criticism very clear: while it might be possible to explain some basic perception and action without invoking mental representations, explaining real, human-level thinking will require representations. This kind of critique has been repeated many, many times in the intervening years. The frequent repetition of Kirsh’s critique (not always with citation) has led many of those who defend radical embodied cognitive science¹ to offer explanations of how one might explain what Clark and

¹ We will use Andy Clark (1997) term as a blanket for all prominent non-representational approaches to cognitive science, including work in the traditions of ecological psychology (Gibson 1966, 1979), situated robotics (Brooks 1991), dynamical systems theory (Port and Van Gelder 1995), and enactivism (Varela et al. 1991).

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Toribio (1994) called representation-hungry behaviors—i.e., real human thinking—without invoking mental representations. Gallagher has recently called this the “scaling-up problem,” i.e., the problem of making current approaches to “lower-order or basic types of processes involving perception and action” also work for explaining “higher-order cognitive abilities, such as memory, imagination, reflective judgment, and so on” (2017, p. 187).

There have been, in general, three kinds of solutions offered by radical embodied cognitive scientists to the scaling-up problem, with most theorists embracing some combination of them. First, several theorists have claimed that human development and participation in cultural settings, including narrative practices and skill learning, lead to new kinds of abilities to tell stories and use cultural artifacts that enable us to have thoughts about things that are distant in space and time (Gallagher and Hutto 2008; Rietveld and Kiverstein 2014). Second, several theorists echo James Gibson (1979) claims about the use of affordances—directly perceivable, environmental opportunities for action—as the source of our abilities to think about things distant in space and time. Humans, and perhaps some other animals, are able to use affordances to imagine future action and to engage in virtual actions. Affordances, the claims is, are already for actions that have not yet occurred, and humans can use them to plan several steps into the future (Gibson 1979; Gallagher 2017; Di Paolo et al. 2017). These first two proposed solutions to the scaling up problem are about our cognitive *abilities*. The third approach concerns the neural *mechanisms* that humans might use in solving representation-hungry problems, with some radical embodied cognitive scientists suggesting that our brains use emulation, or re-presentation of sensory information (Grush 2004), to think about things that are far away (Thompson 2007) and others preferring predictive processing models (Friston 2009) in which the main activity of our brains is the prediction of future experiences (Bruineberg and Rietveld 2014).

It is against this backdrop that the claims of this paper are made. We offer two proposals. First, we suppose that some combination of affordance-based virtual action and development within a human culture (i.e., the first two kinds of solution to the problem of scaling up) does offer the right explanation of our higher-order cognitive capabilities. What we propose here is a very general theory about what we are doing when we engage those abilities. For example, whatever the source of the ability, we are capable right now of wondering whether Shaun Gallagher, who is not here, would like this article, which he might read at some time in the future. The first of our proposals, which we call *artifactualism*, is a story about what we are doing when we have explicit thoughts like this. We propose that what we are doing is *building tools* to help us deal with possible futures in which Shaun Gallagher reads this article. Although we will rely on work in recent philosophy of science concerning models to make our case, our proposal is very much in line with ideas from James (1907), Vygotsky (1978), and Ryle (1949, 1968). Our second proposal, which we call *neural resonance*, is for a possible process which might instantiate artifactualism in brain-body-environment systems. This proposal does conflict with mechanisms (emulation, predictive processing) that have already been proposed as solutions for the scaling-up problem. We will argue that neural resonance is empirically plausible and is better suited

to the interests of radical embodied cognitive scientists than are emulation and predictive processing.

Before moving to the main part of the paper, it is important to be clear about the relationship between these two proposals. Artifactualism is a very general, philosophical theory about the nature of thinking; neural resonance is an empirical theory about brains in bodies in environments that, if correct, could support artifactualism. Artifactualism could still be true even if neural resonance is empirically disconfirmed or otherwise fails as a scientific theory. With that said, the paper is structured as follows. Our artifactualist proposal is that to think is to build and use a tool, and Sect. 2 gives a general account of this, inspired by pragmatist approaches in philosophy of science. Section 3 describes neural resonance, a specific proposal of what the creation and (re)use of thoughts as tools can look like in brain-body-environment systems. Section 4 is a brief conclusion.

2 Artifactualism about thinking

2.1 A day in the life

You are at the breakfast table, drinking coffee. You look at the clock and realize you are late for work. You stand up and pat your pocket to see if you have your keys. You don't, so you wonder where they are. You usually leave them on your nightstand, so you decide to look there first. You walk toward the bedroom, and on the way there you see the laundry pile and think that you need to do laundry tonight, which reminds you that you don't like the smell of the new laundry detergent you bought recently, which makes you wonder what detergent Satoshi uses that makes Satoshi's clothes always smell so good. You get to the bedroom, see your keys on the nightstand and grab them. On your way to the front door you notice the dirty dishes. You consider loading the dishwasher, but you decide you can't do that right now, so you grab your bag to get ready to leave. As you grab your bag, you remember you are going to need a certain book today, so you open the bag to see if the book is in there. Inside the bag you see your laptop and you notice, on the laptop case, the mashed core of an apple you ate the day before. While you bemoan being such a slob, you grab the apple core and throw it away in the kitchen. You wet some paper towels, bring them to your bag and wipe your laptop case. You figure it looks clean enough, close your bag and rush out the door having forgotten to get the book. As you walk out the door you see the emptied trash cans on the curb and you think about getting them, but you immediately rule that out: you can't deal with that right now or you will be late...

The story above illustrates the way our explicit thoughts typically come up in daily contexts: it is in the midst of our various activities that we think, reflect, understand, realize, decide, plan, and remember. It is in the midst of what we do and see that we think about what we do and see as well as about what is not present then and there—what we might see and do later, or what we have in the past. You see the clock and realize you are late, and you see the pile of dirty clothes and make plans to do laundry tonight: in both cases there is something in your surroundings

Table 1 Example of a sequence of explicit, occurrent thoughts and the perception–action events they are part of

Perception–action	Explicit, occurrent thoughts
Drinking coffee. See clock	“8:15! Late for work!”
Stand up. Pat pants pocket	“ <i>Keys? No keys!</i> ” “ <i>Where are they?</i> ” “ <i>Usually I leave them on my nightstand</i> ”
Walk toward bedroom. Pass laundry pile	“I need to do laundry tonight” “ <i>I don’t like the way the new detergent smells. Satoshi always smells so good. I wonder what kind he uses.</i> ”
Continue into bedroom, see keys on nightstand. Grab keys	“There!”
Walk toward door. See dirty dishes	“Nope. Too late.”
Grab bag	“ <i>I need that Edith Stein book. Where is it?</i> ”
Look in bag. See laptop. Smell, then see mashed apple core	“Ewww!” “ <i>Why am I such a slob? I ruin everything.</i> ”
Grab apple core. Go to kitchen. Fling apple in sink. Wet paper towel. Wipe laptop case	“Good enough, I guess.” “I’m late!”
Go out door. See emptied trash cans	“I can’t deal with that now.”
...	...

Thoughts in italics are *not* directed at objects that are perceptually present (e.g., wondering where your keys are)

that guides your thinking. But thinking about something present can also lead us to thinking about things that are *not* present: this happens when you look for an object (e.g., trying to find your keys in the process of preparing to leave) and also when you reflect on your situation and imagine counterfactual ones (e.g., thinking about the laundry detergent you have right now and then thinking about other options and wondering what detergent someone else uses). Table 1 gives a breakdown of the story above emphasizing this intimate connection between thoughts and specific perception–action events, also highlighting how thoughts relate to objects that are perceptually available or not. To be clear, in listing perception–action and thoughts separately, we do not mean to suggest that the two are independent and distinct in kind: on the contrary, thoughts are always associated with our active embodied coping with the world. Still, listing thoughts separately highlights the fact that even thoughts about objects that are not immediately available for perceptual inspection (e.g., the missing keys, on the nightstand in another room) can be useful for coping with aspects of the situation that *are* perceptually present (e.g., looking at the clock and realizing you’re late). Also to be clear, we wish to emphasize that although all of explicit, occurrent thoughts in Table 1 are linguistic, we do not think that all explicit, occurrent thoughts are cases of “talking to oneself”, nor do we think that all explicit, occurrent thoughts are linguistically structured. In the rest of Sect. 2 of this paper, we will draw analogies between thinking and scientific modeling, and just as there are many kinds of scientific models, there are many kinds of explicit, occurrent thought.

2.2 Artifactualism about scientific models

The account we want to offer of *thinking* is inspired by a view that has grown in popularity in the recent philosophy of science literature on models and simulations, according to which the various types of *models* scientists use—including concrete scale models, mathematical equations and computer simulations—are best understood as *tools, instruments or artifacts* (see, e.g., Morrison and Morgan 1999; Keller 2000, 2002; Knuuttila 2011, 2017; Isaac 2013; Currie 2017; Sanches de Oliveira under review). Traditional philosophical analysis has focused on explaining how models *represent* target real-world phenomena: influential discussions about models that prioritize some notion of representation include those by van Fraassen (1980, 2008), Giere (1988, 2004, 2010), Suárez (2004, 2015), Godfrey-Smith (2006), Chakravartty (2010), Pincock (2012), Weisberg (2012), Morrison (2015), and Frigg and Nguyen (2017) among many others (see an overview in Sanches de Oliveira 2018). But artifactualists like the ones mentioned above question this traditional representationalist focus, and they propose that the task of elucidating how models represent target phenomena is secondary, and perhaps even unnecessary, compared to the task of understanding *how models are used*. This artifactualist shift corresponds to a shift away from analysing scientific models as “*models of*” toward understanding models as “*models for*”.

Particular scientific models are *for some action* in that they are designed to be manipulated in specific ways and to accomplish specific tasks. This “what for” of models can be given a thinner or thicker description.² Consider the Phillips Hydraulic Computer, a machine comprising of pipes and tanks through which water flows, that was used in the 1940s and 50s at the London School of Economics to model the flow of money in the economy and relations such as the ones between savings, investment, and consumption. A thinner description of this machine might be that it is for observing and intervening in the water volume of various tanks by manually adjusting how much water flows through certain pipes. A thicker description, by contrast, might be that the Phillips machine is for testing hypotheses about the dynamics of money flow, for explaining past events (e.g., a rise in investments or a drop in spending), for predicting the effect of certain interventions (e.g., changes in taxation), and so on. Besides *hypothesis testing*, *explaining past events* and *predicting future events*, models might more generally be for a variety of different purposes under a thick description, such as *generating understanding*, *enabling interventions*, *designing new experiments*, *simplifying complex problems*, and *unifying apparently disparate phenomena*. What is important to notice is that both thinner and thicker descriptions are important if you want to understand what a model is for, just as they are important if you want to understand what any tool is for. A hammer is for

² Here we follow the distinction introduced by Gilbert Ryle (1968) between “thin” descriptions of behaviors/events and “thick” descriptions of the same behaviors/events. Ryle gives helpful examples: we can describe the same eyelid contraction as “blinking” and as “winking,” or the same hand movement as “inscribing letters on paper” and as “signing a peace treaty and ending a war”; in both pairs, the first description is thinner and the second is thicker, but both are different descriptions of the same event rather than descriptions of different events.

gripping the handle and for swinging so as to strike another object, and at the same time the hammer is for driving a nail in and hanging a picture on the wall—both constitute what the hammer is for, and they are not different things one does when hammering, but are the same act described in different ways.

Importantly, models are also *for someone* insofar as they are designed to be used by scientists with specific skills and interests, and to contribute to inquiry in specific research contexts. The Lotka–Volterra equations, for example, are widely used for modeling predator–prey relations, but they are more useful for the biologist who studies population dynamics than for the biologist who seeks to understand how particular predator and prey species relate to each other given, say, their comparative anatomy and physiology. Similarly, the Khepera robot has been useful for some modelers studying cooperative behavior in ants (Krieger et al. 2000) and phonotaxis in crickets (Reeve et al. 2005), yet it is of little use for the entomologist interested in the molecular genetics of ants or crickets. These cases show that the “whom for” and the “what for” of models are inseparable: if a computer simulation system like the Global Forecast System is for making predictions of future weather states, this is only possible for researchers with access to the right infrastructure, data and technical know-how. These examples from scientific modeling mirror the case of ordinary tools: if we say that knives are for cutting, for example, it is implied that they are for animals capable of prehensile grip. Emphasizing the user in this way highlights how models, as tools, are always evaluated in terms of both their *usefulness* and their *usability*. Models have to be functional and effective toward their “what for,” that is, they have to produce the intended results and “get the job done” (Isaac 2013). But models also have to be at least minimally workable and user-friendly: whether it “gets the job done” or not depends not only on the intended end-goal, but also on the intended user and on the user’s ability to put the model to work. Thus, understanding models as “models for” naturally motivates thinking about models as tools that are used by someone to do something in some context.

Philosophers of science have found this pragmatist understanding of scientific models as “tools for” attractive for a number of reasons, two of which we find particularly helpful for present purposes. The first is that artifactualism helps to shed light on how models and modeling techniques travel across scientific contexts and disciplines (see, e.g., Sanches de Oliveira 2016, under review). Models often turn out to be useful beyond their initially intended domain: this has been the case, for example, in the application of game-theoretical models (from economics) and of nonlinear dynamical mathematical models (from physics) in biology and cognitive science. Artifactualism frames this capacity in terms of the versatility that tools in general have, where function can change in a range of ways depending on the context and the user: for instance, chairs are for sitting on, but you can also step on a chair to change a light bulb, and you can also lay down to sleep on a chair if you are a small dog. In scientific and ordinary cases there can be shifts in what and whom the tool is for, but the tool’s success will still be best understood in relation to some use and some user.

Second, as proponents of artifactualism about models suggest, artifactualism helps to make sense of the presence of idealization and abstraction in model-based science (see, e.g., Isaac 2013; Sanches de Oliveira under review). Philosophers of science

increasingly recognize that, as far as representing goes, models can be pretty lousy: models are never perfect copies of their targets because they always neglect various details and also posit additional elements not found in the target phenomena—modeling always involves some degree of abstraction and idealization, even “rampant and unchecked idealization” (Potochnik 2017). This has led many to ponder and wonder just how these “misrepresentations,” “falsehoods” or “fictions” can generate better, more explanatory models (see, e.g., Wimsatt 1987; Elgin 2004; Bokulich 2012; Kennedy 2012; Morrison 2015; Potochnik 2017). Note, however, that the epistemic role of abstraction and idealization is only puzzling if we think, in traditional representationalist fashion, that models are justified as “models of” some targets, and are justified only to the extent that they (accurately) represent targets. For artifactualists, abstraction and idealization need not be seen as processes of misrepresentation; moreover, falling short from complete and accurate representation of some target (given some definition of representation) need not be seen as a shortcoming. If models are “models for”—for certain uses and for certain users—then any process that makes the model more useful and more usable will, by definition, make it a better model: and this is so even if, on some or other account of representation, these processes are properly described as making the model less representationally accurate.

To conclude, we want to emphasize that, for artifactualists, the idea that models are evaluated in terms of their usefulness and usability does not entail a negation of the representational status of models. There are many different accounts of the nature of the representation relation, and, depending on which definition of representation you choose, the same model may or may not be properly described as representing some target. Artifactualism does not claim that scientific models *never* meet formal definitions of representation. Instead, the claim is that meeting formal definitions of representation is *incidental* to the purpose of models. Ultimately, it does not matter if a model “represents” some target (e.g., if Barbara Webb’s robotic cricket represents a real cricket): what matters instead is the role that the model plays in the activities, interventions, and understanding of scientists. For artifactualists, models are always “models for”: they are tools scientists design, construct and manipulate to accomplish various goals; and, as such, models are more or less valuable depending on how engaging with the model and understanding it leads to success in a range of activities in a given scientific research context.

2.3 Thoughts as tools

The artifactualism about thinking we want to propose in this paper is a straightforward application of artifactualism about scientific models to the domain of the mind. In our view, *explicit, occurrent thoughts* are *tools, instruments, or artifacts* that some agent creates and uses. Mirroring the artifactualist view of models as “models for” rather than “models of”, we propose that thoughts are best understood as being “thoughts for” rather than “thoughts of”. The purpose of thinking is *not* to accurately represent the world: thinking is rarely if ever a matter of aimlessly taking stock of the elements present in our surroundings and generating or updating

an internal inventory of the world. Thoughts are *for something*, they are tools for action, tools that are useful for accomplishing some goal.

The morning routine story narrated in 2.1 illustrates how thoughts can be useful for coping with a number of different perception–action demands. The thoughts that you are late and that you need to find your keys are useful tools for organizing your current actions and for adapting to your present circumstances (e.g., understanding that you need to go look for your keys before you can leave the house). Making plans such as deciding to take care of the laundry, dishes and trash only later in the day is making tools for organizing future actions as well as current ones: having those thoughts can guide what you do at other points throughout the day (e.g., deciding to skip happy hour to go home and deal with the mess, or deciding to go to happy hour to avoid the mess), and it also informs what you do right now insofar as it enables you to draw your attention back to what is currently most pressing, namely getting ready and leaving for work. Besides *coping with our current situation* and *planning future actions*, thoughts can be tools that we create and use for a variety of different purposes, including *simplifying complex problems*, *predicting future experiences*, *explaining past experiences*, *categorizing experiences*, *confirming or disconfirming expectations*, *enabling adaptivity*, *generating understanding*, and more.

Thoughts are also *for some user*; they are tools created for *someone* to use. This point might seem trivial: unlike models, which are public objects that can be interacted with by different users, thoughts are typically private creations of individuals. What we want emphasize here is that, as tools, thoughts are not simply more or less *useful* for some end, but they are also more or less *usable* for someone. The thoughts you entertain cannot just be useful in some abstract and impersonal way: you need to know how *you* can use them. Our ordinary thoughts tend to satisfy this condition, and this is because, generally speaking, we are successful perceiver-actors: contrary to what parts of the psychological literature might seem to suggest, perceptual illusions and motor failures are the exception, and we typically get by quite well. But we do sometimes have thoughts that we fail to use appropriately, and even some we do not know how to use well. For an example of failure to use a thought appropriately, notice that in the narrative from Sect. 2.1, the thought about the needed Edith Stein book did not lead to an appropriate action concerning the book. Possible examples of thoughts we do not know how to use well are some of our thoughts concerning morality. Although we thankfully never find ourselves in true trolley problem-like moral dilemmas, we may occasionally face minor morally difficult situations. Suppose you think a colleague is in the wrong, but you do not have a clear sense of what you can do about it, say, because what your colleague did was a retaliation against something inappropriate someone else had done first. The judgment that what your colleague did was wrong might generally be a useful tool, but in this case it is not straightforwardly usable. Other mundane examples of thoughts that might not be useful to particular users can be found everywhere in traditional schooling. Learning abstract concepts, theories, facts and so on can be understood as the creation (for oneself) of tools that are useful and, over time, becoming a proficient user by making the tool usable. Consider, for example, a mnemonic like RESE NESE MRMN SRSR (it reads as “Rese, Nese, Mr. Man, Sir Sir”). This is a useful tool for remembering how the endings of German articles and adjectives change to agree with gender and

number (masculine, feminine, neuter, and plural) in each of the declensions (nominative, accusative, dative, and genitive). If you, reader, are not a learner of German, the example will not make any sense, which illustrates our point that the pragmatic success of thoughts is relative to the user as much as it is to the goal or use it is for.

In the previous subsection we briefly discussed two advantages of artifactualism as an approach to model-based science. The advantages were, first, that artifactualism helps make sense of the versatility of models and how they come to be successful in different research contexts across scientific disciplines, and second, that artifactualism does away with the mystery surrounding the epistemic role that idealization and abstraction play in modeling. These two points translate into benefits of *artifactualism about thinking* as well.

First, consider how thoughts, as tools, can be (and often are) used beyond the specific situations in which we first create them. As we pointed out above, devising a plan to do laundry at the end of the day helps you deal with your present situation (i.e., making sure you leave for work without getting even later than you already are) and the same thought can turn out to be useful at a later time in a different place (e.g., when you are leaving work and decide not to meet friends at the bar). Of course, the reuse of thoughts is not always positive and helpful. The thought you had in the morning that you are a slob could, at a later time, turn out to be useful for helping you get your act together. Given common normative societal standards, this would probably be a good thing. But you might also use that tool for ends that are not as positive given the same standards: dwelling on negative thoughts could lead to generally poorer performance and weaker personal relationships, reinforcing an unhappy downward spiral. Importantly, the evaluation of these two uses as positive and negative, respectively, makes sense only in relation to some given normative standard. From a pragmatic standpoint these are just different uses to which the tool can be put—that is, both are examples of what the tool is useful for—and thus, the two examples illustrate how thoughts, as tools, are versatile and often turn out to be useful for different purposes in different contexts. This is the case even if we have reasons to prefer some purposes over others.

The second advantage mirrors the usefulness of idealization and abstraction in scientific modeling. Here, the point is that thoughts can be useful even when they do not represent the world completely and accurately on any account of representation. Notice how, if we were to give a representational account of thinking, many of the thoughts that we create and use actually misrepresent the world. In the story we told earlier, when you see the big pile of dirty clothes and tell yourself you are going to do laundry tonight, this thought might accurately capture something about the current state of affairs (i.e., the clothes in the pile *are* dirty and need washing). Yet, there is an important sense in which the thought you had is *not* a truth-evaluable representation of the world because, insofar as it is a plan, it is about something that does not yet exist (i.e., the future, what you might do tonight). But this is not limited to prospective thinking. Confabulation and misremembering are additional

examples of phenomena that support our account.³ Work in various research traditions and domains reveals that inaccurate remembering is not necessarily a pathological symptom, but can also help healthy adults preserve their sense of well-being, identity, and meaning, and can even contribute to their epistemic agency (see, e.g., Bortolotti and Cox 2009; Bortolotti 2018; Conway and Loveday 2015; Gantman et al. 2017; Goldney et al. 2009; Howe 2011). This suggests that the primary purpose of remembering is not to give a faithful recapitulation of past events, but rather to orient our current activities. Examples of prospective and retrospective thinking thus illustrate how, under a representationalist rubric, many of our thoughts would be properly described as involving idealizations and abstractions that misrepresent reality. In practice, however, our thoughts are ultimately evaluated in terms of how useful (adaptive, fitness-enhancing, conducive to health and well-being, epistemically beneficial, etc.) they are.

Artifactualism about thinking proposes that the purpose of explicit, occurrent thoughts is *not* to accurately represent the world. That is, according to artifactualism, standard semantic notions like reference or content are not essential features of explicit, occurrent thoughts. But it is important to realize that artifactualism is *not* committed to the elimination of aboutness. On the contrary, aboutness plays a crucial role in the pragmatist view of thoughts as “thoughts for,” and here we borrow from a Heideggerian understanding of tools to briefly suggest how this is possible.

Heidegger thought that tools are thoroughly referential. Tools are never meaningless chunks of matter floating in a void and which we have to reflectively bestow meaning upon. Rather, we typically encounter tools in the midst of our active, purposive engagements with the world and, as such, tools are typically *about* many things in many interrelated ways. First and foremost, a tool (or “equipment”) is always “something in-order-to” and, in this sense, a tool is *about what it is for*, it is *about* the practice (or “work”) it is a part of. Second, and for this reason, a tool is always *about other tools* that also constitute that same practice. Heidegger affirms: “Taken strictly, there ‘is’ no such thing as *an* equipment. To the Being of any equipment there always belongs a totality of equipment, in which it can be this equipment that it is” (Heidegger 1962/2001, p. 97). A pen is for writing, but a specific kind of writing, one that requires ink and paper (rather than say, light and a computer screen). And, as such, a pen “can be the equipment that it is” only in the totality of equipment that make up (this specific way of) writing—that is, a pen only makes sense as a pen because of ink and paper, and, in this sense, a pen is always about (or “refers to”) ink and paper, even in their absence (Kaufer and Chemero 2015). Third, in addition to being about the totality of equipment, tools are also *about* the materials they are made of: “Hammer, tongs, and needle, refer in themselves to steel, iron, metal, mineral, wood, in that they consist of these” (Heidegger 1962/2001, p. 100). And, lastly and most importantly, tools are always *about* ourselves as *users*. Heidegger claims: “The work produced refers not only to the ‘towards-which’ of its usability and the ‘whereof’ of which it consists: under simple craft conditions it also has an assignment to the person who is to use it or wear it” (Heidegger 1962/2001,

³ We thank Marek McGann for this suggestion.

p. 100). Heidegger explains that in the production of standardized goods the reference may be fixed to some average user rather than a specific person, but the referential character is still there. A tool is, thus, about us in that it is for *us* to do something with it, something that is “for-the-sake-of” and determined by the “totality of our involvements” (Heidegger 1962/2001, p. 116), or our “care structure.” In sum, as a commentator has put it, for Heidegger “every useful thing refers, variously, to other useful things, activities, projects, goals and roles” (Cerbbonne 2008, p. 45).

We think that these various referential aspects of tools found in a Heideggerian account provide us with a particularly insightful way to make sense of the aboutness of thinking. Following the Heideggerian account, thoughts, as tools, can be *about* many things, including ourselves and what we are trying to accomplish. Ultimately, however, thoughts are tools for action, tools that we create, use and evaluate based on how useful and usable they are—based on their “expediency” rather than their “agreement” with reality, as William James put it.⁴ This does not mean that explicit, occurrent thoughts *never* meet formal definitions of representation. Instead, our claim is that meeting formal definitions of representation is *incidental* to the purpose of explicit, occurrent thoughts. It does not matter if our thoughts “represent” some target (e.g., if your judgment that you are a slob or your plan to do laundry later accurately and fully correspond to reality): what matters instead is the role that those thoughts play in your activities, interventions, and understanding. Thoughts are best understood as “thoughts for” rather than “thoughts of”: thoughts are tools we create and use to accomplish various goals and, as such, they are more or less valuable depending on the extent to which they contribute to our successful coping with the world.

3 Ecological psychology and the resonant brain

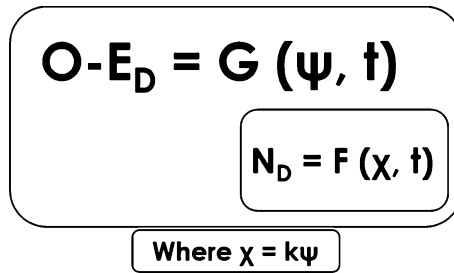
Artifactualism about thinking, as we have proposed in the previous section, is a very general account of the nature and purpose of explicit, occurrent thoughts. Here we will provide a more detailed instantiation of our general account based on nonlinear dynamics and ecological psychology. While we are convinced that this will be a fruitful way to cash out our general artifactualist view, we recognize that the detailed instantiation to follow is speculative. Yet, potential criticisms of this detailed instantiation should not be seen as automatically undermining artifactualism as a whole—the general view we proposed above can stand on its own and should remain a live option even if the specific instantiation we propose here turns out to be problematic. We present the following proposal as a means to show that artifactualism about thinking can be spelled out in empirically rich and consequential ways, subject to testing in the laboratory.

⁴ Our account also coincides with suggestions made by other pragmatists, such as Richard Rorty: “When the contemplative mind, isolated from the stimuli of the moment, takes large views, its activity is more like deciding what to *do* than deciding that a representation is accurate” (Rorty 1980, p. 723).

Ecological psychology is a theory of perception and action first developed by James Gibson (1966, 1979; see also Chemero 2009). There are three main tenets to the theory. First, perception is for action and action is (also) for perception. Perception and action are two sides of the same coin: we perceive to control our action and we need to act in order to access the information we use to perceive the world around us—e.g., we need to locomote to generate the optic flow where the information for locomotion control is available. For this reason, the second main tenet of ecological psychology is that perception is of opportunities for action or *affordances* (Turvey 1992, Chemero 2003). Due to the deep connection between perception and action, the objects of perception are action-relevant offerings of the environment and not the classic physical properties of objects, such as size or color. And last but not least, the third main tenet of ecological psychology is that perception is direct. Ecological psychologists reject the classic argument of the poverty of stimulus (Michaels and Carello 1981), according to which environmental stimulation is not informative enough as to control action and a mechanism of enrichment of such stimulation (e.g., a computational mechanism that builds up a representation of the environment) must be postulated for perception—and other psychological processes—to occur. According to ecological psychologists, environments offer sufficient information about what they afford (ecological information), so a mechanism for the enrichment of stimulation to make it an informative representation of the environment is not needed. This is the sense in which perception is direct: because the stimulus is not impoverished, there are no internal states of the agent (i.e., representations) that mediate her perceptual relation with the environment.

A relevant aspect of ecological psychology for our project is that it does not offer an account of explicit thinking (see, e.g., Sanches de Oliveira and Raja 2018). As a theory of perception and action, ecological psychology deals very well with perception–action events, such as, for example, guiding our pace when we run a marathon or controlling our arm and hand when we grasp a glass of wine. However, many of the events we refer to as “thinking” seem to deal with objects and information that are not directly available in the environment in the current moment. For example, we remember the name of the main character of that novel we like or we plan what we have to buy in the grocery store after work. Thus, in the terms of ecological psychology, the main question regarding “thinking” is how we are able to deal with situations that depend upon information that is not currently available in the local environment.⁵ Our artifactualist proposal to answer this question is that we deal with the information that is not available in the environment in a given moment in a very

⁵ Recall our note above that explicit, occurrent thinking is not limited to linguistic activity. With that reminder, we wish to point out that there is a research program closely related to ecological psychology that attempts to account for linguistic activity, both linguistic interaction with others and speaking silently to oneself: ecolinguistics (Cowley and Steffenson 2015; Cowley and Harvey 2016; Rączaszek-Leonardi et al. 2018). Ecolinguistics is about language and linguistic thinking as embodied activities. The proposal in this section concerns the role that the central nervous system (CNS) plays in activities such as these. We endorse ecolinguistics and wish to be clear that what we say about resonance does not compete with it. Indeed, we think that resonance can serve as an account of the CNS’s role in the activities that ecolinguistics studies..



The diagram illustrates an abstract model of resonance. It consists of a large rounded rectangle containing two smaller rounded rectangles. The top-left rectangle contains the equation $O-E_D = G(\psi, t)$. The bottom-right rectangle contains the equation $N_D = F(\chi, t)$. Below these two rectangles, centered, is a small rounded rectangle containing the text "Where $\chi = k\psi$ ".

Fig. 1 Abstract model of resonance. In a given psychological event, the organism–environment dynamics ($O-E_D$) can be defined as a function, G , of ecological information, ψ , through time, t . The CNS dynamics during the same event (N_D), can be defined as a function, F , of some variable, χ , through time, t . The key aspect of resonance, as defined by Raja (2018), is that $O-E_D$ and N_D are coupled in terms of ecological information such that $\chi = k\psi$

similar way to how we deal with the information that is available in the environment in a given moment. Or, in the jargon of ecological psychology, we *resonate* to both kinds of information.

Resonance is the central metaphor used by Gibson and later ecological psychologists to refer to the activity of the central nervous system (CNS) in perceptual events. In a famous passage, Gibson says:

Instead of supposing that the brain constructs or computes the objective information from a kaleidoscopic inflow of sensations, we may suppose that the orienting of the organs of perception is governed by the brain so that the whole system of input and output resonates to the external information. (1966, p. 5).

It is our contention that we can develop a detailed instantiation of the artifactualist approach to thinking by expanding this idea of resonance.

In a basic, physical sense, resonance is a phenomenon that occurs when a system A is driven by a system B to oscillate at greater amplitude at one or more specific frequencies. This phenomenon is present, for example, when a person is pushing a swing. Only when the swing is pushed at a specific (resonant) frequency—i.e., at specific moments in its cycle—does the amplitude of its swinging increase. Otherwise, the pushing interferes with the swing’s motion. Another instance of resonance may be seen when a struck tuning fork causes another tuning fork nearby to vibrate. As in the case of the swing, the driven tuning fork only resonates to the driving one when their resonant frequencies are related. A further example of a resonant system is the acoustic guitar. The body of acoustic guitars resonates to a played string as both share a resonant frequency or a range of resonant frequencies.

The physical notion of resonance illustrates the feature of the phenomenon that is interesting for the cognitive sciences: when two systems enter into a resonant state, their dynamics are coupled in terms of some variable; for example, the two systems vibrate together (tuning forks or acoustic guitars), or are mechanically synchronized (the “pusher” and the swing). In this sense, resonance is useful in cognitive science and, in particular, in ecological psychology to refer to events of coupling, phase-locking, synchronization, etc. Raja (2018) has defined the *ecological notion*

of resonance as the activity by which the dynamics of the interaction between an organism and its environment (the organism–environment system) are coupled to the dynamics of the organism’s CNS in terms of ecological information (see Fig. 1).

This notion of resonance fits well with both the main tenets and the basic scope of ecological psychology. It is a notion of resonance that makes sense of perceptual events, where on-line ecological information constrains the dynamics of the organism–environment systems and, in so doing, constrains the dynamics of the CNS. In this sense, it departs from the strictly physical notion of resonance while maintaining the features that make it an interesting concept for the cognitive sciences in the first place (e.g., coupling, synchronization). It is worth noting, however, that such an ecological notion of resonance may be reconciled with the use of the same concept at the physiological scale, which is gaining attention in neuroscience in recent years. For example, in the literature on mirror neurons the appeal to motor resonance is common (Gangitano et al. 2004, Borroni et al. 2011; Press et al. 2011; Carvallo et al. 2013, McCabe et al. 2014). Also, different forms of resonance have been described at the scale of neural networks both as an element that contributes to the overall functioning of the nervous system—e.g., single-neuron resonance (Hutcheon and Yarom 2000; Kasevich and LaBerge 2011), resonance frequency shift (Lau and Zochowsky 2011; Shtrahman and Zochowski 2015), coherence resonance (Yu et al. 2018), network resonance (Helfrich et al. 2019), or stochastic resonance (Ikemoto et al. 2018)—and as a resource that participates in different cognitive functions such as learning (Roach et al. 2018). It is far from the scope of this paper to further analyze the relationship between the ecological and physiological notions of resonance, but such a relationship is possible in principle and will deserve more attention in future work (see Raja 2019).

As we have already noted, the proposed ecological notion of resonance deals with on-line information. However, an important part of explicit thinking—and of “high” cognition in general—occurs with no available ecological information concerning the target of the thought, which might be distant in space and/or time. Thinking entails, for example, remembering previous events or anticipating future ones and, therefore, it sometimes has targets that are no longer present or that are not present yet. Thus, the main question for ecological psychology regarding thinking may be reformulated as concerning whether resonance is able to accommodate events when some of the information needed is not available in the local organism–environment dynamics. If it can, resonance might be able to serve as a way to describe the activity of the CNS in both thinking and perception–action events.

The basic phenomenon of resonance is simple: a driving system affects the dynamics of a driven system at particular frequencies—i.e., affecting its oscillatory properties. However, resonance can be very complex in the real world. The dynamics of systems, driving ones or driven ones, are not always smooth and steady. Many times, those dynamics are changing, chaotic, or noisy. Moreover, the interactions between systems are usually very complex and consist of multiple, different systems engaged in multiple, different patterns of interaction. Such complexity in the dynamics and the interactions between resonant systems makes them exhibit a set of important common properties: spontaneous oscillation, entrainment, and nonlinearity

(Large and Kolen 1994; Large 2008, 2010).⁶ According to Large, systems that exhibit nonlinear resonance, along with spontaneous oscillation and entrainment, exhibit interesting behaviors: they act like filters for their resonant frequency; they resonate to frequencies that are not present in the stimulation they receive; and they anticipate the dynamics of the driving system. Neural oscillators also spontaneously oscillate at different frequencies and instantiate different degrees of coupling (or entrainment) to the systems they interact with. In addition, neural oscillators exhibit nonlinear resonance and intrinsically nonlinear dynamics, which makes them relate to each other in terms of different patterns of coupling.⁷ The fact that neural oscillators exhibit these properties has two consequences. First, nonlinear resonance is a plausible physical and mathematical characterization of neural oscillators. Second, nonlinear resonance is a plausible explanation of how a system composed of neural oscillators, like the CNS, can act as a filter, anticipate future events, and respond to signals that are not present.

Due to spontaneous oscillation, neural oscillators exhibit dynamics with specific rhythms at some frequencies (Guckenheimer and Kuznetsov 2007). These dynamics and their frequencies may be equal to or different from the frequencies of other systems they interact with, but, when the different systems interact, their dynamics (frequencies, amplitude of oscillation, etc.) affect each other. This phenomenon is what Large refers to as entrainment. Interestingly, when entrainment occurs, the synchronization between two systems (e.g., driving and driven, or an organism-environment event and a neural oscillator) is perfect when their frequencies are similar. However, when the frequency of the driven system (e.g., neural oscillator) is greater than that of the driving system (e.g., organism-environment event), the dynamics of the driven system anticipate the dynamics of the driving one (Large 2008, p. 203, Fig. 6.5B).⁸ Finally, in their interactions, driving and driven systems are coupled to each other in different patterns. For example, they can couple in terms of their main frequencies, but also in terms of the harmonics of those frequencies. This is what makes them non-linear resonators: their patterns of resonance are not always linearly related to the main frequency of the stimulation. It is true that driven systems tend to resonate to frequencies near to their own resonant frequency (this is how they act as filters), but they also resonate to the harmonics of the main frequency of the driving system. Moreover, the harmonics of the driven systems also resonate to the harmonics of the driving one. And, even more, driven systems remain coupled to those harmonics even when the driving system is not acting as stimulation anymore (Large 2008, pp. 203–204, Fig. 6.5C). In this sense, they resonate to non-current stimulation or, in other words, to absent dynamics of the driving system.

⁶ Sometimes also referred as *higher order resonance* (e.g., Large 2008, p. 203 & ff.).

⁷ We deliberately avoid all the mathematics that support these claims. In case those details are of interest for the reader, the work of Edward Large is very complete in terms of the mathematics and modelling of neural oscillators and non-linear resonance (Large and Kolen 1994, Large 2008, 2010; Large et al. 2016).

⁸ This kind of anticipation is fully compatible with the one described in the ecological literature (Stepp and Turvey 2010; Stepp et al. 2010). In this sense, Large's model of neural oscillators as non-linear resonators might serve as an example of the way resonance accounts for ecological anticipation.

The three features of neural oscillators as nonlinear resonators make resonance a promising candidate to account for the activity of the CNS not only in perceptual events, but also explicit thinking. The three features of non-linear resonators allow for dealing with information (filtering) that is not present in the moment the thought is happening, both future information (anticipation) and past information (non-current stimulation). This is the kind of information that we deal with in “high” cognition, so if resonance can account for the way the CNS deals with it, then it is also a promising way to describe the activity of the CNS as a component of the activity of thinking. To be clear, we are not claiming that all thinking can be explained by the sole appeal to a relatively simple notion such as resonance. Rather, what we are proposing here is that a theory built upon the notion of resonance could provide an interesting model to better understand thinking.

To this point, we have suggested that resonance may be an ecological way to understand the activity of the CNS in explicit thinking as far as it deals with future and past information. However, resonance—specifically nonlinear resonance—also makes it possible to understand thinking in the artifactualist way defended in the previous section. On the one hand, as we have seen above, nonlinear resonance allows for filtering. In the case of an agent-environment interaction, there is an overwhelming amount and variety of environmental information available. Nonlinear resonance in the CNS can act as a filter, selecting the right environmental information for successful action. For example, having a thought like “I need to find the red book” might help someone explore a bookshelf with many books in a more effective way. Resonance could account for what is going on in the CNS while the agent is having such a thought during an ongoing action.

Moreover, as nonlinear resonators are able to anticipate future states of driving systems, they may help agents cope with the environment in terms of its future states. In this sense, the activity of the CNS while having a thought like “I need to have water during the long flight” that serves to modulate ongoing action, such as looking for a place to buy water at the airport, could be explained in terms of such an anticipation enabled by nonlinear resonance. And finally, nonlinear resonance allows for sensitivity to information (e.g., environmental states) that is not currently available. This is often what explicit thinking is for. And because it enables the use of information that is no longer in the local environment, nonlinear resonance can account for the role of the CNS regarding this characteristic of thinking as well. A thought like “I left my car in the third floor of the parking lot” modulates the current action by making the agent sensitive to information that is not in the current environment. A possible mechanism for this sensitivity is the CNS resonating to such nonlocal information.

There are several ways in which the artifactualist approach to thinking is compatible with resonance. Consider, first, the fact that, according to artifactualism, thoughts act as tools by modulating the ongoing perception–action dynamics, and that resonance offers a way to describe the dynamics of the CNS when explicit thinking acts, for example, as a filter, as an anticipatory tool, or as a “memory” tool. It is important to notice that the appeal to explicit thinking does not exhaust the explanation of the overall dynamics of the agent-environment interaction, nor does the appeal to resonance exhaust the explanation of explicit thinking. As we have

seen, thoughts appear as tools during ongoing actions (see Table 1) and, as such, they are components of the overall action of the agent in her environment. Similarly, nonlinear resonant states of the CNS are components of the activity of thinking. They are the activity of the CNS during thinking, but do not exhaust the explanation of thinking itself—i.e., nonlinear resonant states only describe the role of the CNS in the tool that a thought is. In this sense, resonance and the artifactualist approach to thinking are compatible, but they explain different things. Artifactualism explains what thinkers are doing; neural resonance explains how their CNS interacts with the rest of the organism-environment system.

Second, through resonance we propose to understand the role that the CNS plays in thinking in terms of its dynamics. Whether or not the CNS meets some formal definition of representation is a secondary matter. In this sense, the understanding of the CNS in terms of resonance may be taken to be agnostic regarding representation. It does not matter if a neural (resonant) state represents some aspect of the environment. What is crucial concerning the activity of the CNS during thinking, and what needs to be explained, is the way it deals with, or resonates to, the relevant information. Neural resonance and artifactualism thus have the very same view on representations. Thus, taking resonance as a promising way to describe the activity of the CNS as a component of the activity of thinking is compatible with artifactualism in that regard as well.

As we noted at the beginning of this section, our proposal concerning neural resonance is a tentative account that makes the artifactualist approach to thinking empirically accessible. It is perfectly possible, however, to accept artifactualism regarding thinking while rejecting resonance as a way to understand the activity of the CNS during thinking. Nevertheless, the idea of resonance helps to bring artifactualism to a place where predictions can be generated and hypotheses can be tested. By taking thinking as partially constituted by resonant neural states, we have the opportunity to use theoretical and empirical tools both from ecological psychology and neural reuse, for example, to explore the relation between thinking, the brain, and organism-environment dynamics, and to analyze the way thoughts affect processes of attention, anticipation, or memory.

4 Conclusion

In this final section, we will draw some explicit connections between our two proposals in this article and prior responses to the scaling-up problem for radical embodied cognitive science. We will also put our proposals in the context of some historical antecedents. To recap, we have proposed what we call artifactualism, a very general view about the nature of explicit thoughts. Explicit thinking is the creation and use of models, and models are to be understood as artifacts that serve action-guiding purposes. We have also proposed a speculative empirical instantiation of artifactualism, namely that explicit thoughts occur when neural and bodily resources resonate with events in brain-body-environment systems.

First, neural resonance is not consistent with either of the neural mechanisms proposed as answers to the scaling-up problem: emulation and predictive processing.

Neural resonance is inconsistent with emulation because emulation is a straightforwardly representationalist notion, developed as a theory of representation from cybernetics by Grush (2004). The basic case of an emulator is a forward model, a kind of representation that can be used to compute the outcomes of actions prior to the arrival of sensory feedback. Whether predictive processing models involve representations is somewhat controversial, with a small minority holding that predictive processing models are not representational (e.g., Bruineberg et al. 2018). This issue aside, neural resonance and predictive processing are likely incompatible because, to date, models based on predictive processing are markedly brain-centric, giving to body and environment a secondary role in the explanation of psychological events. Neural resonance, by contrast, is based on the ecological tradition and aims to be a story about the CNS as it is constrained by body and environment. In this sense, and unlike what can be observed in the literature on predictive processing, in terms of resonance, the brain is a blue-collar brain (Van Orden et al. 2012). That is, rather than being the executive in charge, the activity of the brain works for, i.e., is controlled by, bodily behavior.

That said, it is important to be clear that our description of the disagreements among neural resonance, emulation, and predictive processing is in no way intended as an argument against these views as empirical research programs. Indeed, it is quite possible that human cognition involves emulation, predictive processing, and neural resonance (and many more neural and bodily actions). At the same time, however, it is clear that emulation is in conflict with artifactualism. While forward models could be viewed as artifacts designed to guide action, they guide action specifically in virtue of representing. If explicit thinking is found to be in general or for the most part emulation, this would count as evidence against artifactualism. On the other hand, finding that emulation is a key part of the neural mechanism for a limited range of human actions (e.g., ballistic movements as Grush has proposed) would not affect artifactualism, which is only a theory of explicit thinking.

As noted above, artifactualism about explicit thinking is fully compatible with the other attempts by radical embodied cognitive scientists to solve the scaling-up problem, i.e., the claims that the abilities to think about things that are not local in space and time arise through development in a cultural context that includes storytelling, virtual actions, and complex tool use. The model building that humans engage in when they think is cultural through and through. This point is essentially Vygotskian. In his work on the ontogenesis of psychological function, Vygotsky (1978) proposed that internal, intrapersonal psychological operations are always a “reconstruction” or “internalization” of external, interpersonal activities. It is only after many months of speaking and using tools with others that children develop the abilities to do so on their own. As Vygotsky claims: “All the higher functions originate as actual relations between human individuals” and “Aspects of external or communicative speech as well as egocentric speech turn ‘inward’ to become the basis of inner speech” (pp. 56–57). Indeed, in many cases, the ability to think of things that are not local is built right into the tools themselves. To repeat an example from Rietveld and Kiverstein (2014), once one is educated in architectural practice, one can use a local scale model to guide actions with respect to a building that will be built in the future, in another city. These abilities are also built into human language. The

point of artifactualism is that explicit thinking is building analogues of scale models, typically, but not always, by talking to oneself. As Ryle put it,

...what *le Penseur* is trying to do in saying things to himself is that he is trying, by success/failure tests, to find out whether or not the things that he is saying would or would not be utilisable as leads or pointers. They are not pointers, but only candidate-pointers; and most of them will have to be turned down after examination. ... [S]o, I suggest, in his pondering, reflecting, deliberating, etc. the thinker is not guiding himself anywhere, but trying to find out whether this or that track of his own making would or would not qualify as a guiding, as opposed to a mis-guiding or non-guiding, track. (Ryle 1968, p. 508)

Artifactualism, however, differs from prior attempts by radical embodied cognitive scientists to solve the scaling-up problem in that it is about the activity of thinking, rather than the necessary precursors that enable that activity. The activity of thinking, according to artifactualism, is a kind of model building. The models we build when we think are tools to help us cope with the situations we are in, and situations we might be in in the future.

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